

Doeblert matrix: a chemometric tool for analytical chemistry—review

Sérgio L.C. Ferreira^{a,*}, Walter N.L. dos Santos^a, Cristina M. Quintella^a,
Benício B. Neto^b, Juan M. Bosque-Sendra^c

^a Instituto de Química, Universidade Federal da Bahia, Salvador, Bahia 40170-290, Brazil

^b Departamento de Química Fundamental, Universidade Federal de Pernambuco, Recife, Pernambuco 50740-540, Brazil

^c Department of Analytical Chemistry, Faculty of Sciences, University of Granada, Fuentenueva s/n, E-18071 Granada, Spain

Received 21 November 2003; received in revised form 21 January 2004; accepted 21 January 2004

Available online 17 March 2004

Abstract

A review of the use of the Doeblert matrix as a chemometric tool for the optimization of methods in analytical chemistry and other sciences is presented. The theoretical principles of Doeblert designs are described, including the coded values for the use of this matrix involving two, three, four and five variables. The advantages of this matrix in comparison with other response surface designs, such as central composite and Box–Behnken, designs are discussed. Finally, 57 references concerning the application of Doeblert matrices in the optimization of procedures involving spectroanalytical, electroanalytical and chromatographic techniques are considered.
© 2004 Elsevier B.V. All rights reserved.

Keywords: Chemometric; Doeblert matrix; Review

1. Introduction

As complexity, variety, growing importance of quality and omnipresent uncertainty mark today's chemistry, a statistical approach to experimental design is almost inevitable. Therefore, the use of multivariate experimental design techniques is becoming increasingly widespread in analytical chemistry and other sciences. Multivariate designs, which allow the simultaneous study of several control variables, are faster to implement and more cost-effective than traditional univariate approaches [1,2]. Several experimental design models exist that reduce the number of experiments and that can be used in different cases. Thus, if it is desired to detect influential factors, experimental designs for first-order models (factorial designs or Plackett–Burman designs) can be used. On the other hand, to approximate a response function or to optimize a process, experimental designs for second-order models should be used.

The most popular first-order design is the two-level full (or fractional) factorial, in which every factor is experimentally studied at only two levels. Due to their simplicity and

relatively low cost, full factorial designs are very useful for preliminary studies or in the initial steps of an optimization, while fractional designs are almost mandatory when the problem involves a large number of factors [3]. However, since only two levels are used, the models that can be fit to these designs are somewhat restricted. Consequently, if a more sophisticated model is required, as for the location of an optimum set of experimental conditions, then one must resort to designs for second-order models (response surface designs), which employ more than two factor levels to allow fitting of a full quadratic polynomial. In analytical chemistry, the response surface designs most used are symmetrical and describe a spherical experimental domain.

A central composite design (CCD) [4] combines a two-level full or fractional factorial design with additional points (star points) and at least one point at the center of the experimental region, selected to obtain properties such as rotatability or orthogonality, in order to fit quadratic polynomials. For fitting quadratic response models, the CCD is a better alternative to the full factorial three-level design because its performance is comparable at a lower cost. Therefore, it has been the most accepted experimental design for second-order models. The total number of design points needed (N) is determined by the formula $N = 2^k + 2k + C_0$, where k is the number of variables and

* Corresponding author. Tel.: +55-71-2355166; fax: +55-71-2355166.
E-mail address: slcf@ufba.br (S.L.C. Ferreira).

Table 1

Comparison of efficiency of central composite design (CCD), Box–Behnken design (BBD) and Doehlert design (DM)

Variables (K)	Number of coefficients (p)	Number of experiments (f)			Efficiency (p/f)		
		CCD	DM	BBD	CCD	DM	BBD
2	6	9	7	–	0.67	0.86	–
3	10	15	13	13	0.67	0.77	0.77
4	15	25	21	25	0.60	0.71	0.60
5	21	43	31	41	0.49	0.68	0.61
6	28	77	43	61	0.36	0.65	0.46
7	36	143	57	85	0.25	0.63	0.42
8	45	273	73	113	0.16	0.62	0.40

C_0 is the number of center points. In analytical chemistry, this design has been widely used.

Box–Behnken designs (BBD) [5] are rotatable second-order designs based on three-level incomplete factorial designs. The special arrangement of the BBD levels allows the number of design points to increase at the same rate as the number of polynomial coefficients. For three factors, for example, the design can be constructed as three blocks of four experiments consisting of a full two-factor factorial design with the level of the third factor set at zero. The number of experimental points (N) is defined by the expression $N = 2k(k - 1) + C_0$, where k is the number of variables and C_0 is the number of center points.

An alternative and very useful experimental design for second-order models is the uniform shell design proposed by Doehlert in 1970 [6]. Doehlert designs are easily applied to optimize variables [7,8] and offer advantages in relation to central composite and Box–Behnken designs. They need fewer experiments, which are more efficient and can move through the experimental domain. Despite these attractive features, however, it took some time until researchers started to take notice of Doehlert designs.

In analytical chemistry, the first application of the Doehlert matrix involved the optimization of a separation process using HPLC [9]. Since then, several papers have appeared involving determinations by spectrometric, chromatographic and electroanalytical methods.

This paper offers a critical discussion of the different chemometric approaches developed for the optimization of chemical and instrumental variables using Doehlert matrices

Table 2

Doehlert matrix for two-variables

	Experimental variables	
	A	B
1	0	0
2	1	0
3	0.5	0.866
4	–1	0
5	–0.5	–0.866
6	0.5	–0.866
7	–0.5	0.866

Table 3

Doehlert matrix for three-variables

	Experimental variables		
	A	B	C
1	0	0	0
2	1	0	0
3	0.5	0.866	0
4	0.5	0.289	0.817
5	–1	0	0
6	–0.5	–0.866	0
7	–0.5	–0.289	–0.817
8	0.5	–0.866	0
9	0.5	–0.289	–0.817
10	–0.5	0.866	0
11	0	0.577	–0.817
12	–0.5	0.289	0.817
13	0	–0.577	0.817

and the different applications of these experimental designs in analytical chemistry and other sciences.

2. Doehlert matrices

The Doehlert design describes a spherical experimental domain and it stresses uniformity in space filling. Although this matrix is neither orthogonal nor rotatable, it does not significantly diverge from the required quality for effective use [7]. For two variables, the Doehlert design consists of one central point and six points forming a regular hexagon, and therefore situated on a circle. In three dimensions it can be viewed in different ways, depending on the geometric

Table 4

Doehlert matrix for four-variables

	Experimental variables			
	A	B	C	D
1	0	0	0	0
2	1	0	0	0
3	0.5	0.866	0	0
4	0.5	0.289	0.817	0
5	0.5	0.289	0.204	0.791
6	–1	0	0	0
7	–0.5	–0.866	0	0
8	–0.5	–0.289	–0.817	0
9	–0.5	–0.289	–0.204	–0.791
10	0.5	–0.866	0	0
11	0.5	–0.289	–0.817	0
12	0.5	–0.289	–0.204	–0.791
13	–0.5	0.866	0	0
14	0	0.577	–0.817	0
15	0	0.577	–0.204	–0.791
16	–0.5	0.289	0.817	0
17	0	–0.577	0.817	0
18	0	0	0.613	–0.791
19	–0.5	0.289	0.204	0.791
20	0	–0.577	0.204	0.791
21	0	0	–0.613	0.791

Table 5
Doehlert matrix for five-variables

	Experimental variables				
	A	B	C	D	E
1	0	0	0	0	0
2	1	0	0	0	0
3	0.5	0.866	0	0	0
4	0.5	0.289	0.817	0	0
5	0.5	0.289	0.204	0.791	0
6	0.5	0.289	0.204	0.158	0.775
7	−1	0	0	0	0
8	−0.5	−0.866	0	0	0
9	−0.5	−0.289	−0.817	0	0
10	−0.5	−0.289	−0.204	−0.791	0
11	−0.5	−0.866	−0.204	−0.158	−0.775
12	0.5	−0.289	0	0	0
13	0.5	−0.289	−0.817	0	0
14	0.5	−0.289	−0.204	−0.791	0
15	0.5	−0.289	−0.204	−0.158	−0.775
16	−0.5	0.866	0	0	0
17	0	0.577	−0.817	0	0
18	0	0.577	−0.204	−0.791	0
19	0	0.577	−0.204	−0.158	−0.775
20	−0.5	0.289	0.817	0	0
21	0	−0.577	0.817	0	0
22	0	0	0.613	−0.791	0
23	0	0	0.613	−0.158	−0.775
24	−0.5	0.289	0.204	0.791	0
25	0	−0.577	0.204	0.791	0
26	0	0	−0.613	0.791	0
27	0	0	0	0.633	−0.775
28	−0.5	0.289	0.204	0.158	0.775
29	0	−0.577	0.204	0.158	0.775
30	0	0	−0.613	0.158	0.775
31	0	0	0	−0.633	0.775

structure selected [10]. In Doehlert designs the number of levels is not the same for all variables. In a two-variable Doehlert design, for example, one variable is studied at five levels while the other is studied at only three levels. This property allows a free choice of the factors to be assigned to a large or a small number of levels. Different criteria can be used to assign the factors. As a general rule, it is preferable to choose the variable with the stronger effect as the factor with five levels in order to obtain most information of the system.

Each design is defined considering the number of variables and the coded values (C_i) of the experimental matrix. The relationship between coded and real values is given by

$$C_i = \left\{ \frac{X_i - X_i^0}{\Delta X_i} \right\} \alpha$$

where C_i is the coded value for the level of factor i , X_i is its real value in an experiment, X_i^0 is the real value at the center of the experimental domain, ΔX_i is the step of variation of the real value and α is the coded value limit for each factor. The number of experiments required (N) is given by $N = k^2 + k + C_0$, where k is the number of variables and C_0 is the number of center points. Replicates at the central level

of the variables are performed in order to validate the model by means of an estimate of experimental variance.

The comparison among the second-order designs cited have demonstrated that Doehlert matrices and Box–Behnken designs are more efficient than composite central designs, considering that the efficiency of one experimental design is defined as the number of coefficients of the model estimated divided by the number of experiments. Table 1 presents a comparison of the efficiency of central composite, Box–Behnken, and Doehlert designs. It shows that for all values of k the Doehlert design is the most efficient of the three. Doehlert designs are also more efficient in mapping space: adjoining hexagons can fill a space completely and efficiently, since the hexagons fill space without overlap [7]. Another advantage is its potential for sequentially [7,11], where experiments can be re-used when the boundaries have not been well chosen at first.

Many of the papers reporting the use of a Doehlert matrix involve the optimization of a process controlled by only two variables, for which seven experiments are required. The coded values of this matrix are given in Table 2. For three variables, the matrix and its dimensions depend of the geometric structure involved [10]. Tables 3–5 contain the coded values of the factor levels for Doehlert designs on three, four and five variables, respectively.

3. Methodological proposals to determine optimal conditions

When the aim of the study is the optimization of the experimental variables, i.e. the determination of the co-ordinates of the stationary point, the first step of the procedure is the analysis of the fitted response surface, to estimate the optimal settings of the variables. There are three methods to assess the response surface.

3.1. Inspection of the response surfaces

In an analytical process where several independent variables ($A, B, \dots$) influence a experimental response, the observed response (R) is a certain function of the levels of the variables, $R = f(A, B, \dots)$. The surface that is represented by this function is called the response surface, which is an n -dimensional surface in the $(n + 1)$ -dimensional space. In order to get a useful graph, a 2D representation of a 3D graph or a contour graph can be drawn. If the function depends on three or more variables, visualization is possible only if one or more variables are set to a constant value. When the response surface can be obtained, inspection of the 2D surface or the corresponding contour diagram allows us to determine the geometrical nature of the surface, e.g. a stationary ridge, a rising ridge, a simple maximum, a simple minimum or a saddle point (minimax) [3]. A saddle point is defined as the stationary point of a surface response which presents the maximum response for the levels of some vari-

ables and simultaneously the minimum response for the levels of other variables of the analytical system studied.

3.2. Canonical analysis

Canonical analysis [3] is a mathematical treatment which consists of changing the origin of the plot from its original co-ordinates to the stationary point and rotating the axes until they correspond to the principal axes of the contours. Using this new co-ordinate system, the second-order model equations are simplified and its geometrical nature becomes apparent.

The algebraic signs of the canonical equation coefficients provide an idea about the nature of its stationary point. If the values are all negative it is a maximum; if all positive, it is a minimum, and if the signs are mixed it is a saddle point.

3.3. Lagrange's criteria

Lagrange's criteria are a mathematical procedure to determine the nature of the stationary point of a function [12].

3.3.1. Lagrange's criterion for two-variable functions

Consider the optimization of a process, which has two experimental variables (A and B) and the experimental response (R). For a second-order model, the experimental data fit the function:

$$R = a + b(A) + c(B) + d(A)^2 + e(B)^2 + f(A)(B) \quad (1)$$

where R is the experimental response to be optimized, a is the constant term, b and c are coefficients of the linear terms, d and e are coefficients of the quadratic terms and f is the coefficient of interaction between the two factors.

Lagrange's criterion is based on the calculation of the Hessian determinant of R , given by

$$H = \begin{vmatrix} \frac{\partial^2 R}{\partial A^2} & \frac{\partial^2 R}{\partial A \partial B} \\ \frac{\partial^2 R}{\partial B \partial A} & \frac{\partial^2 R}{\partial B^2} \end{vmatrix}$$

which reduces to

$$H = \left(\frac{\partial^2 R}{\partial A^2} \right) \left(\frac{\partial^2 R}{\partial B^2} \right) - \left(\frac{\partial^2 R}{\partial A \partial B} \right) \left(\frac{\partial^2 R}{\partial B \partial A} \right)$$

The critical point, $(A, B) = (A_0, B_0)$, is a maximum if $H(A_0, B_0) > 0$ and $\partial^2 R / \partial A^2(A_0, B_0) < 0$, and a minimum if $H(A_0, B_0) > 0$ but $\partial^2 R / \partial A^2(A_0, B_0) > 0$. A saddle point exists if $H(A_0, B_0) < 0$.

3.3.2. Lagrange's criterion for three-variable functions

The second-order model for three variables (A , B and C) is given by

$$R = a + b(A) + c(B) + d(C) + e(A)^2 + f(B)^2 + g(C)^2 + h(A)(B) + i(A)(C) + j(B)(C)$$

where R is the experimental response to be optimized, a is the constant term, b , c and d are coefficients of the linear terms, e , f and g are coefficients of the quadratic terms and h , i , and j are coefficients of interaction between the three factors.

If the quadratic function only shows one stationary point (A_0, B_0, C_0) , four situations are possible:

1. There is not any information: $\Delta_2 = 0$.
2. Relative maximum: $\Delta_1 < 0$; $\Delta_2 > 0$; $\Delta_3 < 0$.
3. Relative minimum: $\Delta_1 > 0$; $\Delta_2 > 0$; $\Delta_3 > 0$.
4. Saddle point: none of the above situations applies.

where Δ_3 is the Hessian determinant of the function $H(A, B, C)$, Δ_2 and Δ_1 are calculated using the following equations:

$$\Delta_1 = \frac{\partial^2 R}{\partial A^2}$$

$$\Delta_2 = \begin{vmatrix} \frac{\partial^2 R}{\partial A^2} & \frac{\partial^2 R}{\partial A \partial B} \\ \frac{\partial^2 R}{\partial B \partial A} & \frac{\partial^2 R}{\partial B^2} \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} \frac{\partial^2 R}{\partial A^2} & \frac{\partial^2 R}{\partial A \partial B} & \frac{\partial^2 R}{\partial A \partial C} \\ \frac{\partial^2 R}{\partial B \partial A} & \frac{\partial^2 R}{\partial B^2} & \frac{\partial^2 R}{\partial B \partial C} \\ \frac{\partial^2 R}{\partial C \partial A} & \frac{\partial^2 R}{\partial C \partial B} & \frac{\partial^2 R}{\partial C^2} \end{vmatrix}$$

Initial information about the geometrical nature of the surface can be obtained from the signs and magnitudes of the quadratic coefficients in the polynomial functions. If all these coefficients are negative, the function can show a maximum; if all these coefficients are positive the function can show a minimum; and when some are positive and the others are negative, the stationary point can be a saddle point with a relative maximum for the variables with a negative term, and with a relative minimum for the ones with a positive coefficient. Nevertheless, it is always necessary to verify these conclusions by applying the Lagrange criteria.

4. Determination of the co-ordinates of the stationary point

The co-ordinates of the stationary point $(A_0, B_0, \dots, C_0)$ are calculated differentiating the response function with respect to each variable and solving the equation system: $\delta R / \delta A = 0$, $\delta R / \delta B$ and $\delta R / \delta C = 0$.

Table 6
Papers involving spectroanalytical techniques optimized by Doehlert matrix

Analyte	Optimization (factors number)	Analytical technique	Sample	References
As	Extraction conditions for speciation (3)	ICP-MS	Mussels	[13]
As	Instrumental conditions for the determination (2)	GFAAS	Petroleum refinery streams	[14]
As	Instrumental conditions for the determination (4)	GFAAS	Naphtha	[15]
Cd	Chemical variables (2)	SPS	Natural water	[16]
Cu	On-line preconcentration system (2)	FAAS		[17]
Fe, V, Mo, Sn	Variables of the solvent extraction process (2)			[18]
Ge	Experimental conditions (3)	SPHM	Drugs	[19]
Li	Spectrophotometric determination (2)	SP-MAS	Drugs	[20]
Mn	On-line preconcentration system (2)	ICP-OES	Biological	[21]
Mo	Instrumental conditions of atomization (3)	GFAAS		[22]
Mo	Preconcentration procedure (2)	ICP-OES	Seawater	[23]
Ni	Preconcentration using cloud point extraction (4)	FAAS	Saline effluents	[24]
Pb	On-line preconcentration system (2)	FAAS	Natural food	[25]
Several metals ions	Liquid–liquid extraction (2)	FAAS	Synthetic samples	[26]
Si	Instrumental conditions for the determination (4)	GFAAS	Naphtha	[27]
V, Cu	Preconcentration procedure (2)	ICP-OES	Seawater	[28]
Zn	On-line preconcentration system (3)	FAAS	Natural water	[29]
Zn	Chemical variables (2)	SPS	Natural water	[30]
Zn	On-line preconcentration system (2)	ICP-OES	Seawater	[31]
Zn, Cd	Chemical and instrumental variables (2)	SPDS	Natural water	[32]

SPHM: spectrofluorometry; SPS: solid phase spectrophotometry; SPDS: solid phase derivative spectrophotometry.

5. Applications in analytical chemistry

5.1. Application of Doehlert matrix in spectroanalytical techniques

The Doehlert matrix has been widely used for optimization of methods involving spectroanalytical techniques, to analyze a wide variety of samples like natural and sea water, mussels, petroleum refinery streams, several kinds of drugs, biological substances and synthetic samples. Usually, the papers explain the optimization of experimental and/or instrumental conditions in procedures of preconcentration and/or separation and determination procedures using fluorimetry, spectrophotometry, GFAAS and ICP-AES. Table 6 describes some characteristics of these papers.

5.2. Electroanalytical techniques

The Doehlert matrix was used several times for the optimization of electroanalytical methods. Procedures using adsorptive stripping voltammetry were optimized for the determination of Nimesulide [33] and Kynurenic acid [34]. The experimental conditions for the determination of quinolinic acid in human plasma and urine samples using differential pulse polarography were also established [35].

5.3. Application of Doehlert matrices in chromatographic techniques

The Doehlert matrix has proved its usefulness on several types of chromatography, where it was used to optimize the separation of species and to increase the selectivity of the methods. This matrix has been used in the systematic and

simultaneous optimization of the gradient solvent system and to optimize instrumental and experimental variables [36] and also for the optimization of the mobile phase in the separation of chlorophenols [37].

It has been applied to the analysis of petroleum and its derivatives using supercritical-fluid chromatography for the separation of aromatic-compounds in petroleum fractions [38]. This matrix has also been used in the analysis of gasoline using capillary gas chromatography [39].

Doehlert designs were used for multicomponent separation of enantiomers using direct capillary gas chromatography [40], for the separation of selected tropane alkaloids by micellar electrokinetic capillary chromatography [41], to optimize separation of oxadiazon residues by headspace solid-phase microextraction and gas chromatography–mass spectrometry [42] and in the detection of organochlorine pesticides in water by gas chromatography and electron-capture detection [43].

This multivariate optimization was also applied to the glucosinolate separation by micellar electrokinetic capillary chromatography [44], to the derivatization of biogenic amines as dabsyl chloride preceding to the determination by RP-HPLC [45] and in the supported liquid membrane extraction of biogenic amines from wine samples prior to determination by liquid chromatography as dabsyl derivatives [46].

5.4. Application of the Doehlert matrix in other sciences

The Doehlert matrix has also been applied to other fields of science, due to its generality and ease of use. It was used to optimize process variables and in formulation and quality control in the development of pharmaceutical prod-

ucts [47,48]. It was also applied to examine virus adsorption [49,50].

In materials science, the use of this matrix produced better yields in different processes, such as quality control of granulation in a high shear mixer [51], conversion coatings on stainless steel in nitric acid solution [52], adsorption of metallic ions onto fly ash [53], preparation of activated carbon from olive-waste cakes [54] and coagulation–flocculation of raw water [55].

This optimization method was also applied to the experimental conditions of enzymatic synthesis [56], in the production of pectate lyase by a recombinant *Escherichia coli* [57], in the improvement of the production of *Penicillium cyclopium* lipase [58] and *P. cyclopium* partial acylglycerol lipase [59]. Also, it was used to determine the effect of factors on the accumulation of trehalose and glycerol production by *Saccharomyces cerevisiae* [60]. Other more eclectic applications were the adsorption process of radioelements on mixtures of minerals [61] and the investigation of the thermal behavior of extractants used for reprocessing nuclear fuel [62].

6. Conclusions

The Doehlert matrix is a good design for response surface methodology because it permits: (i) estimation of the parameters of the quadratic model, (ii) building of sequential designs, (iii) detection of lack of fit of the model and (iv) use of blocks. Also, the Doehlert design as a chemometric tool presents advantages over other, more used, response surface designs, such as composite central and Box–Behnken. Doehlert designs, despite their relatively scarce diffusion, are amply used in analytical chemistry and other sciences. Therefore, different applications of these designs in the optimization of procedures involving several analytical techniques are presented in this paper.

References

- [1] D.C. Montgomery, Design and Analysis of Experiments, 4th ed., Wiley, New York, 1997.
- [2] B.B. Neto, I.S. Scarminio, R.E. Bruns, Como Fazer Experimentos, Unicamp, Campinas, 2001.
- [3] G.E.P. Box, W.G. Hunter, J.S. Hunter, Statistics for Experimenters, Wiley, New York, 1978.
- [4] G.E.P. Box, K.B. Wilson, J. R. Statist. Soc., B 13 (1951) 1.
- [5] G.E.P. Box, D.W. Benhken, Technometrics 2 (1960) 195.
- [6] D.H. Doehlert, Appl. Statist. 19 (1970) 231.
- [7] D.L. Massart, B.G.M. Vandeginste, L.M.C. Buydens, S. de Jong, P.J. Lewi, J. Smeyers-Verbeke, Handbook of Chemometrics and Quality Metrics, Part A, Elsevier, Amsterdam, 2003.
- [8] M. Nechar, M.F. Molina, L. Cuadros Rodríguez, J.M. Bosque-Sendra, Anal. Chim. Acta 316 (1995) 185.
- [9] Y. Hu, D.L. Massart, J. Chromatogr. A 485 (1989) 311.
- [10] A.M. García-Campaña, L. Cuadros Rodríguez, A. Lupiañez González, F. Alés Barrero, M. Román Ceba, Anal. Chim. Acta 348 (1997) 237.
- [11] J.M. Bosque-Sendra, M. Nechar, L. Cuadros Rodríguez, M.F. Molina, Anal. Proc. 32 (1995) 375.
- [12] M.J. Valderrama, Métodos matemáticos aplicados a las Ciencias Experimentales, Pirámide, Madrid, 1989, p. 290.
- [13] T. Dagnac, A. Padro, R. Rubio, G. Rauret, Anal. Chim. Acta 364 (1998) 19.
- [14] R.J. Cassella, O.D. de Sant'Ana, R.E. Santelli, Spectrochim. Acta, Part B 57 (2002) 1967.
- [15] M.V. Rebouças, S.L.C. Ferreira, B.B. Neto, J. Anal. At. Spectrom. 18 (2003) 1267.
- [16] M. Nechar, M.F. Molina, J.M. Bosque-Sendra, Anal. Chim. Acta 382 (1999) 117.
- [17] S.L.C. Ferreira, M.A. Bezerra, W.N.L. dos Santos, B.B. Neto, Talanta 61 (2003) 295.
- [18] D. Gazquez, M. Sanchez-Viñas, M.G. Bagur, G. García, J. Anal. At. Spectrom. 13 (1998) 105.
- [19] A.M. García-Campaña, F. Ales Barrero, A. Lupiañez González, M. Román Ceba, Anal. Chim. Acta 447 (2001) 219.
- [20] L. Gámiz Gracia, L. Cuadros Rodríguez, M. Román Ceba, Talanta 44 (1997) 75.
- [21] M. Zougagh, A. García de Torres, J.M. Cano Pavón, Anal. Lett. 36 (2003) 1115.
- [22] Z. Benzo, P. Araujo, A. Sierraalta, F. Ruetter, Anal. Chem. 65 (1993) 1107.
- [23] S.L.C. Ferreira, H.C. dos Santos, M.S. Fernandes, M.S. de Carvalho, J. Anal. At. Spectrom. 17 (2002) 115.
- [24] M.A. Bezerra, A.B. Conceição, S.L.C. Ferreira, Anal. Bioanal. Chem. 378 (2004) 798.
- [25] S.L.C. Ferreira, W.N.L. dos Santos, M.A. Bezerra, V.A. Lemos, J.M. Bosque-Sendra, Anal. Bioanal. Chem. 375 (2003) 443.
- [26] M. Camino, M.G. Bagur, M. Sanchez-Viñas, D. Gazquez, R. Romero, J. Anal. At. Spectrom. 16 (2001) 638.
- [27] J.A.A. Amaro, S.L.C. Ferreira, J. Anal. At. Spectrom. 19 (2004) 246.
- [28] S.L.C. Ferreira, A.S. Queiroz, M.S. Fernandes, H.C. dos Santos, Spectrochim. Acta, Part B 57 (2002) 1939.
- [29] W.N.L. dos Santos, C.M.C. Santos, S.L.C. Ferreira, Microchem. J. 75 (2003) 211.
- [30] M.F. Molina, M. Nechar, J.M. Bosque-Sendra, Anal. Sci. 14 (1998) 791.
- [31] M. Zougagh, P. Cañada Rudner, A. García de Torres, J.M. Cano Pavón, J. Anal. At. Spectrom. 15 (2000) 1589.
- [32] J.M. Bosque-Sendra, M. Nechar, M.F. Molina, Mikrochim. Acta 134 (2000) 43.
- [33] S. Furlanetto, S. Orlandini, G. Aldini, R. Gotti, E. Dreassi, S. Pinzauti, Anal. Chim. Acta 413 (2000) 229.
- [34] S. Furlanetto, S. Pinzauti, P. Gratteri, E. La Porta, G. Calzaroni, J. Pharm. Biomed. Anal. 15 (1997) 1585.
- [35] S. Furlanetto, S. Pinzauti, E. La Porta, A. Chiarugi, P. Mura, S. Orlandini, J. Pharm. Biomed. Anal. 17 (1998) 1015.
- [36] P. Araujo, Trends Anal. Chem. 19 (2000) 524.
- [37] B. Bourguignon, F. Marcenac, H.R. Keller, P.F. Deaguier, D.L. Massart, J. Chromatogr. 628 (1993) 171.
- [38] R. Fraile, V. Sanchez, J. High Resol. Chromatogr. 16 (1993) 169.
- [39] I.P. Durand, S. Gautier, E. Robert, M.C. Guilhem, R. Phan-Tan-Luu, J. High Resol. Chromatogr. 20 (1997) 289.
- [40] J. Krupčík, M. Greša, I. Špánik, E. Benická, J. Hrouzek, I. Skačáni, P. Sandra, J. Chromatogr. A 779 (1997) 253.
- [41] L. Mateus, S. Cherkaoui, P. Christen, J.L. Veuthey, J. Chromatogr. A 829 (1998) 317.
- [42] A. Navalón, A. Prieto, L. Araujo, J.L. Vilchez, J. Chromatogr. A 946 (2002) 239.
- [43] C. Aguilar, A. Peñalver, E. Pocurull, J. Ferré, F. Borrull, R.M. Marcé, J. Chromatogr. A 844 (1999) 425.
- [44] L. Paugam, R. Ménard, J.P. Larue, D. Thouvenot, J. Chromatogr. A 864 (1999) 155.

- [45] R. Romero, M.G. Bagur, M. Sanchez-Viñas, D. Gazquez, *Chromatographia* 51 (2000) 404.
- [46] R. Romero, Ja. Jonsson, D. Gazquez, M.G. Bagur, M. Sanchez-Viñas, *J. Sep. Sci.* 25 (2002) 584.
- [47] D. Voinovich, M. Moneghini, B. Perissutti, J. Filipovic-Grcic, I. Grabnar, *Int. J. Pharm.* 203 (2000) 235.
- [48] C. Sánchez-Lafuente, S. Furlanetto, M. Fernández-Arévalo, J. Alvarez-Fuentes, A.M. Rabasco, M.T. Faucci, S. Pinzauti, P. Mura, *Int. J. Pharm.* 237 (2002) 107.
- [49] F. Quignon, A. Huyard, L. Schwartzbrod, F. Thomas, *J. Virol. Meth.* 68 (1997) 33.
- [50] F. Quignon, F. Thomas, C. Gantzer, A. Huyard, L. Schwartzbrod, *Water Res.* 32 (1998) 1222.
- [51] D. Vojnovic, D. Chicco, H. El Zenary, *Int. J. Pharm.* 145 (1996) 203.
- [52] S.E. Hajjaji, M.-T. Maurette, E. Puech-Costes, A. Guenbour, A. Ben Bachir, L. Aries, *Surf. Coat. Technol.* 110 (1998) 40.
- [53] P. Ricou-Hoeffer, I. Lecuyer, P. Le Cloirec, *Water Res.* 35 (2001) 965.
- [54] A. Baçaoui, A. Yaacoubi, A. Dahbi, C. Bennouna, R. Phan-Tan-Luu, F.J. Maldonado-Hodar, J. Rivera-Utrilla, C. Moreno-Castilla, *Carbon* 39 (2001) 425.
- [55] M. Franceschi, A. Girou, A.M. Carro-Diaz, M.T. Maurette, E. Puech-Costes, *Water Res.* 36 (2002) 3561.
- [56] A. Ismail, M. Linder, M. Ghoul, *Enzyme Microb. Technol.* 25 (1999) 208.
- [57] C.G. Hounsa, J.M. Aubry, H.C. Dubourguier, J.P. Hornez, *Appl. Microbiol. Biotechnol.* 45 (1996) 764.
- [58] G. Vanot, V. Deyris, M.-C. Guilhem, R. Phan-Tan-Luu, L.-C. Comeau, *Appl. Microbiol. Biotechnol.* 57 (2001) 342.
- [59] G. Vanot, D. Valérie, M.-C. Guilhem, R. Phan-Tan-Luu, L.-C. Comeau, *Appl. Microbiol. Biotechnol.* 60 (2002) 417.
- [60] F. Carvalheiro, J.C. Roseiro, F.M. Gírio, *Food Microbiol.* 16 (1999) 543.
- [61] P. Jacquier, P. Meier, J. Ly, *Appl. Geochem.* 16 (2001) 85.
- [62] T. Dagnac, J.-M. Guillot, P.L. Cloirec, *J. Anal. Appl. Pyrolysis* 37 (1996) 33.